POWER MINIMALITY AND EXCLUSION OF PROPER SURJECTIVE SUBDIRECT D-GROUPOIDS UNDER DIFFERENTIAL GALOIS HYPOTHESES

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Allowed standard prerequisites (editorial).

- Picard-Vessiot theory of linear connections and its Galois correspondence. Author Section 2.3; exact source bibliography retained
- Isogeny theorem for centerless parallelisms. Author Theorem thm:0, citing bc2017 Corollary 4.3
- Goursat lemma for simple Lie algebra subdirect products. Named in author proof of thm:7

1. Preliminaries

We recall the necessary background on $\mathcal{D}$-groupoids, foliations, parallelisms, and Galois theory of fields with operators, and we refer the reader to Malgrange [19], Casale and Blázquez-Sanz [5], Casale [9], Białynicki-Birula [1], and [10] for details. Throughout this section, let M be a smooth irreducible algebraic complex variety of dimension n.

1.1. **$\mathcal{D}$-groupoids or algebraic Lie pseudogroups.** A *Lie pseudogroup* is a pseudogroup (a set of local diffeomorphisms closed under composition, inversion, restriction, and gluing of compatible maps) that is defined by a system of PDE's satisfying some regularity conditions. In the algebraic geometry framework, one can drop the regularity condition and the singular version of Lie pseudogroups will be called *$\mathcal{D}$-groupoids*. By definition, a $\mathcal{D}$-groupoid acting on an algebraic variety M is an algebraic Lie pseudogroup on some dense Zariski open subset of M. There are several equivalent formalizations of this idea; here we will follow the exposition given in [7, Section 2], which is more convenient for our applications as it makes systematic use of the principal structure of the frame bundles.

1.1.1. ***k*-frame bundles and *k*-jet groupoids.** Let M be a smooth complex and irreducible algebraic variety of dimension n. A k-frame r_x in M is a k-jet of a local biholomorphism $\big(\mathbb{C}^n,0\big)\to M$ defined around $0\in\mathbb{C}^n$; its base point $x\in M$ is the image of 0. The space of k-frames forms a principal bundle $R_kM\to M$ modeled over the algebraic group Γ_k of k-jets of local automorphisms of $\big(\mathbb{C}^n,0\big)$ that naturally acts on R_kM by composition on the right side. Note that R_kM is a smooth algebraic variety of complex dimension $n\binom{n+k}{n}$, and Γ_k is an affine algebraic group of complex dimension $n\big(\binom{n+k}{k}-1\big)$. The field $\mathbb{C}(R_kM)$ is referred to as the field of rational differential functions of order $\leq k$. The term *differential* here expresses that the functions in R_kM may depend on the derivatives of functions in M with respect to the Cartesian coordinates $\varepsilon_1,\dots,\varepsilon_n$ in $\big(\mathbb{C}^n,0\big)$.

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Example 1.1. Let $\mathbb{C}^n = M$ with coordinates $x_1, \dots, x_n$. Then, a k-frame is the k-jet $j_0^k r$ of a local biholomorphism $r\colon (\mathbb{C}^n, 0) \to M$, $r(\varepsilon) = (X_1(\varepsilon), \dots, X_n(\varepsilon))$, where the functions $X_j(\varepsilon)$ are assumed to be, without loss of generality, polynomials of degree $\leq k$

$$X_j(\varepsilon) = \sum_{0 \leq |\alpha| \leq k} \frac{1}{\alpha!} \frac{\partial^{|\alpha|} X_j}{(\partial \varepsilon)^\alpha}(0) \varepsilon^\alpha.$$

By defining $x_{j:\alpha}(r) = \frac{\partial^{|\alpha|} X_j}{(\partial \varepsilon)^\alpha}(0)$ we obtain a system of $n\binom{n+k}{k}$ coordinates $\{x_{j:\alpha}\}_{|\alpha| \leq k}$ on $R_k\mathbb{C}^n$; where $x_{j:\alpha}$ represents the α-th derivative of x_j with respect to $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$. By the inverse function theorem, $R_k M$ identifies with the complement in $\mathbb{C}^{n\binom{n+k}{k}}$ of the hypersurface defined by $\{\det\left(x_{j:\epsilon_j}\right) = 0\}$, that is, the vanishing of the Jacobian. Here ϵ_j represents the unit vector in $\mathbb{N}^n$ with j-th coordinate equal to 1. The field of rational differential functions of order $\leq k$ is $\mathbb{C}(x_{j:\alpha} \mid |\alpha| \leq k)$.

Simultaneously, we may consider the groupoid $\mathrm{Aut}_k(M)$ of k-jets of local biholomorphisms defined around any base point in M. This set has the natural algebraic structure of a smooth variety of dimension $n + n\binom{n+k}{k}$. Each element $\sigma_{xy} \in \mathrm{Aut}_k(M)$ has a source $x \in M$ and a target $y \in M$, and two of them may be composed if their respective target and source points coincide, so that $\mathrm{Aut}_k(M)$ has a natural groupoid structure, moreover, Faà di Bruno's formula[1] shows that $\mathrm{Aut}_k(M)$ is an algebraic groupoid (meaning in the category of schemes over $\mathbb{C}$).

Example 1.2. As in Example 1.1, let $\mathbb{C}^n = M$ with coordinates $x_1, \dots, x_n$. An element of $\mathrm{Aut}(M)$ is $j_x^k\sigma$, the k-jet of a local biholomorphism $\sigma\colon (M, x) \to M$, $x + \varepsilon \mapsto \sigma(x + \varepsilon)$ is also given by $\sigma(x_1 + \varepsilon_1, \dots, x_n + \varepsilon_n) = (Y_1(\varepsilon), \dots, Y_n(\varepsilon))$, where the $Y_j(\varepsilon)$ are polynomials of degree $\leq k$ of the coordinates

$$Y_j(\varepsilon) = \sum_{0 \leq |\alpha| \leq k} \frac{1}{\alpha!} \frac{\partial^{|\alpha|} Y_j}{(\partial x)^\alpha}(0) \varepsilon^\alpha.$$

By defining $y_{j:\alpha}(\sigma) = \frac{\partial^{|\alpha|} Y_j}{(\partial x)^\alpha}(0)$, we obtain a system of $n + n\binom{n+k}{k}$ coordinates $\{x_j, y_{j:\alpha}\}_{|\alpha| \leq q}$ in $\mathrm{Aut}(\mathbb{C}^n)$, where $y_{j:\alpha}$ represents the α-th derivative of y_j with respect to $x = (x_1, \dots, x_n)$. The field of rational functions in $\mathrm{Aut}_k\left(\mathbb{C}^n\right)$ is $\mathbb{C}(x_j, y_{j:\alpha} \mid |\alpha| \leq k)$.

However, we can arrive at the algebraic structure of $\mathrm{Aut}_k(M)$ by an alternative path, taking into account that any pair of frames $j_0^k r_x$, $j_0^k r_y$ can be composed to produce a k-jet of biholomorphism $j_x^k\sigma = j_x^k\left(r_y \circ r_x^{-1}\right)$. Any k-jet of biholomorphism can be recovered as such a composition in a unique way, up to an element of Γ_k, which allows us to reconstruct $\mathrm{Aut}_k(M)$ as the quotient by the diagonal action $\mathrm{Aut}_k(M) \simeq (R_k M \times R_k M)/\Gamma_k$.

This is the groupoid of gauge isomorphisms of fibers of the principal bundle $R_k M \to M$. Each k-jet of biholomorphism $j_x^k\sigma$ can be interpreted as a Γ_k-equivariant map from $(R_k M)_x$ to $(R_k M)_{\sigma(x)}$, where the application of $j_x^k\sigma$ to a k-frame r_x is just its composition $j_0^k(\sigma \circ r_x)$, which yields a k-frame at y.

Now, we may consider Zariski closed subgroupoids of $\mathrm{Aut}_k(M)$, but that notion would not be enough to allow us to define $\mathcal{D}$-groupoids. Instead, we will consider Zariski closed subsets that are groupoids *on the generic points*.

Definition 1.3. A rational subgroupoid of $\mathrm{Aut}_k(M)$ is a Zariski closed subset $\mathcal{G} \subset \mathrm{Aut}_k(M)$ such that

(a) There is an open subset $M' \subset M$ such that $\mathcal{G}|_{M'} = \mathcal{G} \cap \mathrm{Aut}_k(M') \mapsto M' \times M'$ is a smooth algebraic groupoid.

[1] Faà di Bruno formula gives explicit expressions for the higher-order derivatives of the composition of functions. See, for instance, [23].

(b) For any open subset $U \subset M$ the intersection $\mathcal{G}|_U = \mathcal{G} \cap \mathrm{Aut}_k(U)$ is dense in $\mathcal{G}$.

There is a fundamental relationship between rational groupoids and algebras of rational functions on R_kM. To each rational groupoid $\mathcal{G} \subset \mathrm{Aut}_k(M)$, we can assign a field of rational differential invariants of order $\leq k$

$$\mathrm{Inv}(\mathcal{G}) = \{f \in \mathbb{C}(R_kM) \mid f(r) = f(\sigma \circ r) \text{ for all } \sigma \in \mathcal{G},\, r \in (R_kM)_{s(\sigma)} \cap \mathrm{dom}(f)\}.$$

Reciprocally, for any rational differential function $f \in \mathbb{C}(R_kM)$, we can define its groupoid of symmetries $\mathrm{Sym}(f)$ as the Zariski closure of the set

$$\{\sigma \in \mathrm{Aut}_k(M) \mid f(r) = f(\sigma \circ r) \text{ for all } r \in (R_kM)_{s(\sigma)} \cap \mathrm{dom}(f)\}$$

and for any set of rational functions $\mathbb{F} \subset \mathbb{C}(R_kM)$, $\mathrm{Sym}(\mathbb{F}) = \bigcap_{f\in\mathbb{F}} \mathrm{Sym}(f)$. Then, as a particular case of [6, Proposition 2.18], we have the following statement.

Proposition 1.4. *The assignments $\mathcal{G} \rightsquigarrow \mathrm{Inv}(\mathcal{G})$ and $\mathbb{F} \rightsquigarrow \mathrm{Sym}(\mathbb{F})$ are inverse bijective correspondences between the sets of rational subgroupoids of $\mathrm{Aut}_k(M)$ and Γ_k invariant subfields of $\mathbb{C}(R_kM)$ containing $\mathbb{C}$.*

For a given rational subgroupoid $(s,t)\colon \mathcal{G} \to M \times M$, there is a Zariski open subset $M' \subseteq M$ such that $\mathcal{G}|_{M'} = (s,t)^{-1}(M' \times M')$ is a smooth algebraic Lie subgroupoid of $\mathrm{Aut}_k(M')$. For $x, y \in M'$, we denote $\mathcal{G}_{xy} = (s,t)^{-1}(\{(x,y)\})$, which is a principal homogeneous space with the right action of the algebraic group $\mathcal{G}_{xx}$ and also with the compatible left action of $\mathcal{G}_{yy}$.

We also denote $\mathcal{G}_{x\bullet} = s^{-1}(\{x\})$, $\mathcal{G}_{\bullet x} = t^{-1}(\{x\})$ and $\mathcal{G}_{xy} = (s,t)^{-1}(\{(x,y)\})$. Then we have that $t\colon \mathcal{G}_{x\bullet}|_{M'} \to M'$ is a $\mathcal{G}_{xx}$-principal homogeneous space where the action is given by the composition.

1.1.2. *$\boldsymbol{\mathcal{D}}$-structure of frame bundles.* The implicit derivation with respect to the standard coordinates in $\big(\mathbb{C}^n, 0\big)$ induces n canonical differential operators, $\delta_i\colon \mathbb{C}(R_kM) \to \mathbb{C}(R_{k+1}M)$.

In order to see these differential operators as derivations of a field, we take the inverse limit $RM = \varprojlim R_kM$ of the sequence

$$\cdots \to R_3M \to R_2M \to R_1M \to R_0M = M$$

as a proalgebraic variety. Elements of RM are formal frames that may or may not be convergent. $RM \to M$ is a principal bundle modeled over the proalgebraic group $\Gamma = \varprojlim \Gamma_k$ of formal automorphisms of $\big(\mathbb{C}^n, 0\big)$. The field of rational functions in RM is

$$\mathbb{C}(RM) = \bigcup_{k\geq 0} \mathbb{C}(R_kM)$$

and thus, the implicit derivative operators become derivations. By setting $\mathcal{D} = \{\delta_1, \ldots, \delta_n\}$ the set of implicit derivative operators, we have that $(\mathbb{C}(RM), \mathcal{D})$ is a $\mathcal{D}$-field in the classical sense of Kolchin.

1.1.3. *Definition of $\boldsymbol{\mathcal{D}}$-groupoid.* Now let us define $\mathrm{Aut}(M)$ as the inverse limit of the sequence

$$\cdots \to \mathrm{Aut}_3 M \to \mathrm{Aut}_2 M \to \mathrm{Aut}_1 M \to \mathrm{Aut}_0 M = M \times M.$$

The elements of $\mathrm{Aut}(M)$ are formal biholomorphisms of M on itself, which may or may not be convergent. By a closed subset in $\mathrm{Aut}(M)$, we mean a closed subset in the initial topology with respect to the projections $\mathrm{Aut}(M) \to \mathrm{Aut}_k(M)$. Such a closed subset is a sequence of Zariski closed subsets $Z = \{Z_k\}_{k\in\mathbb{N}}$ such that

(a) $Z_k \subseteq \mathrm{Aut}_k M$.
(b) Z_{k+1} dominates Z_k by projection.

Definition 1.5. A closed subset $\mathcal{G} = \{\mathcal{G}_k\}_{k\in\mathbb{N}} \subset \mathrm{Aut}(M)$ is a $\mathcal{D}$-groupoid of transformations of M if and only if

(a) For all k, $\mathcal{G}_k$ is a rational subgroupoid of $\mathrm{Aut}_k M$.
(b) For all $f \in \mathrm{Inv}(\mathcal{G}_k)$ and $\delta \in \mathcal{D}$ we have $\delta f \in \mathrm{Inv}(\mathcal{G}_{k+1})$.

This means that if $\mathcal{G}$ is a $\mathcal{D}$-groupoid and we consider the field of invariants of any order

$$\mathrm{Inv}_{\mathcal{D}}(\mathcal{G}) = \bigcup_{k\geq 0} \mathrm{Inv}(\mathcal{G}_k),$$

condition (b) in Definition 1.5 is equivalent to the assertion that $\mathrm{Inv}_{\mathcal{D}}(\mathcal{G})$ constitutes a $\mathcal{D}$-field. We have a correspondence again, between $\mathcal{D}$-groupoids and differential fields, given in [7, Proposition 2.16].

Proposition 1.6. *The assignment $\mathcal{G} \rightsquigarrow \mathrm{Inv}_{\mathcal{D}}(\mathcal{G})$ is a bijective correspondence between the set of $\mathcal{D}$-groupoids of transformations of M and Γ-invariant $\mathcal{D}$-subfields of $\mathbb{C}(RM)$ containing $\mathbb{C}$.*

From Kolchin theorem [15], any differential subfield of a finitely generated differential field is finitely generated as a differential field. In particular, there exists k such that $\mathrm{Inv}_{\mathcal{D}}(\mathcal{G})$ is generated by $\mathrm{Inv}(\mathcal{G}_k)$ as a differential field. This k is the order of $\mathcal{G}$, and thus $\mathcal{G}$ is completely determined by $\mathcal{G}_k$.[2]

1.1.4. *$\boldsymbol{\mathcal{D}_M}$-structure of $\mathbf{Aut}(\boldsymbol{M})$ and its Kolchin topology.* In the original approach to $\mathcal{D}$-groupoid by B. Malgrange [18], the definition is different. We give an overview of Malgrange definition and refer to [11] for the proofs.

This alternative definition makes use of the $\mathcal{D}_M$-structure of $\mathrm{Aut}(M)$. Here, by $\mathcal{D}_M$ we understand the noncommutative ring of differential operators in $\mathbb{C}(M)$ spanned by the Lie algebra of rational vector fields $\mathfrak{X}(M)$. The main point here is that any vector field in M can be extended to a total derivative operator that differentiates functions of $\mathrm{Aut}_k(M)$ into functions of $\mathrm{Aut}_{k+1}(M)$. If X is a rational vector field in M, then

$$X^{\mathrm{tot}}\colon\ \mathbb{C}(\mathrm{Aut}_k(M)) \to \mathbb{C}(\mathrm{Aut}_{k+1}(M)), \qquad \big(X^{\mathrm{tot}} f\big)\big(j_x^{k+1}\sigma\big) = \vec{X}_x\big(f \circ j^k\sigma\big).$$

In particular, derivations that are well defined in some $\mathcal{O}_M(U)$ will give us derivations of the ring of regular functions in $\mathrm{Aut}_k(U)$, this being an open subset of $\mathrm{Aut}_k(M)$.

For the sake of simplicity, let us assume that M is an affine space, there are n regular vector fields linearly independent at each point of M, that is, the module of derivations of $\mathcal{O}_M$ is free of rank M. The general case should be worked out locally.

An ideal $\mathcal{J} \subset \mathcal{O}_{\mathrm{Aut}}(M)$ is a $\mathcal{D}_M$-ideal if for every function $f \in \mathcal{J}$ the total derivatives of f are also in $\mathcal{J}$. A subset $Z = \{Z_k\}_{k\in\mathbb{N}} \subset \mathrm{Aut}(M)$ is Kolchin closed if it is the set of zeros of a $\mathcal{D}_M$-ideal. Kolchin closed subsets are also termed $\mathcal{D}_M$-varieties. An alternative, and historically original [18], definition for $\mathcal{D}$-groupoid goes as follows. *A $\mathcal{D}$-groupoid in M is a $\mathcal{D}_M$-variety $\mathcal{G} = \{\mathcal{G}_k\}_{k\in\mathbb{N}} \subseteq \mathrm{Aut}(M)$ such that for all k, $\mathcal{G}_k$ is a rational subgroupoid of $\mathrm{Aut}_k(M)$.*

1.1.5. *Lie algebra of a $\boldsymbol{\mathcal{D}}$-groupoid.* There is a unique way to extend a vector field X on M, seen as a derivations of $\mathbb{C}(M)$, to a Γ_k invariant derivation of each $\mathbb{C}(R_kM)$ that commutes with the canonical $\mathcal{D}$-structure $\mathcal{D} = (\delta_1, \ldots, \delta_n)$. The extension $X^{(k)}$ is completely determined by the equations $X^{(k+1)}\delta_j I = \delta_j X^{(k)} I$. The flow of the vector field $X^{(k)}$ in R_kM codifies the flow induced on the space of frames R_kM

[2]The system of partial differential equations $\mathcal{G}_k \subset \mathrm{Aut}_k(M)$ determining $\mathcal{G}$ is known in the mathematical literature as a Lie equation [16, 21].

by the flow of X on M. Moreover, the vector field $X^{(k)}$ is Γ_k-invariant. The dynamical interpretation of $X^{(\infty)}$ is just the flow induced by the flow of X on the frame space, in the sense

$$X^{(k)}\big(j_0^k r_x\big) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} j_0^k(\exp(tX)\circ r_x).$$

In the product $R^kM \times R^kM$ we may consider the vector field $\big(0, X^{(k)}\big)$ that represents $X^{(k)}$ acting on the second component. This vector field is Γ_k-invariant, and it has a projection by the canonical quotient map $R^kM \times R^kM \to \mathrm{Aut}_k(M)$, $\big(j_0^k r_x, j_0^k r_y\big) \mapsto j_x^k\big(r_y \circ r_x^{-1}\big)$. We define $\widetilde{X}^{(k)}$ as its projection on $\mathrm{Aut}_k(M)$. The dynamical interpretation of $\widetilde{X}^{(k)}$ is the post-composition with the flow of X

$$\widetilde{X}^{(k)}\big(j_x^k\sigma\big) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} j_x^k((\exp tX)\circ\sigma).$$

Let us consider the linear bundle $J^k(TM/M) \to M$ of k-jets of vector fields. The value of $\widetilde{X}^{(k)}$ at $e_x = j_x^k\mathrm{id}$ depends only on the k-jet of X at x. Therefore, we have a well-defined regular map

$$\iota_k\colon\ J^k(TM/M) \to T(\mathrm{Aut}_k(M)), \qquad j_x^k X \mapsto \widetilde{X}^{(k)}(e_x).$$

This map identifies the space $\big(J^k(TM/M)\big)_x$ with $T_{j_x\mathrm{Id}}(\mathrm{Aut}(M)_{x\bullet})$.[3]

However, this prolongation process can be carried out to infinite order, so that from a vector field X in M we obtain a Γ-invariant derivation $X^{(\infty)}$ of the field $\mathbb{C}(RM)$ and a derivation $\widetilde{X}^{(\infty)}$ of the field $\mathbb{C}(\mathrm{Aut}(M))$. Then, we have the limit

$$\varprojlim \mathrm{Lie}(\mathcal{G}_k) = \mathrm{Lie}(\mathcal{G}) \subset J(TM/M),$$

which is, on the generic point of M, an involutive linear system of partial differential equations in the linear bundle TM/M and a bundle of Lie algebras.

Definition 1.7. For a $\mathcal{D}$-groupoid $\mathcal{G}$, we denote by $\mathrm{Lie}(\mathcal{G}_k) = \iota_k^{-1}(T\mathcal{G}_k) \subset J^k(TM/M)$, and by $\mathrm{Lie}(\mathcal{G}) = \varprojlim \mathrm{Lie}(\mathcal{G}_k) \subset J(TM/M)$.

$\mathrm{Lie}(\mathcal{G}_k) = \iota_k^{-1}(T\mathcal{G}_k)$ is an algebraic family vector spaces $\mathrm{Lie}(\mathcal{G}_k) \to M$ which at the generic point is a vector bundle over M whose rank is the dimension of $\mathcal{G}_k$ minus n. The fiber of $\mathrm{Lie}(\mathcal{G}_k)$ at x is the vector space of k-jets of vector fields X having the property

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} j_x^k(\exp tX) \in T(G_k)_{x\bullet}.$$

That is, the flow of X near x gives a local parametric group of transformations which is *tangent to* $\mathcal{G}_k$.

Note that the k-jet at x of a Lie bracket $[X,Y]$ depends on the $(k+1)$-jets at x of the vector fields X and Y. Therefore, we do not obtain a bundle of Lie algebras, our Lie bracket takes values on jets of smaller order

$$\mathrm{Lie}(\mathcal{G}_{k+1}) \times_M \mathrm{Lie}(\mathcal{G}_{k+1}) \to \mathrm{Lie}(\mathcal{G}_k), \qquad \big(j_x^{k+1}X, j_x^{k+1}Y\big) \mapsto j_x^k[X,Y].$$

However, this bracket defines a Lie bracket on fibers of $\mathrm{Lie}(\mathcal{G})$ called the fiberwise bracket. The term $\mathcal{D}$-Lie algebra will be used for $\mathrm{Lie}(\mathcal{G})$ endowed with the fiberwise bracket. As a Lie algebroid, $\mathrm{Lie}(\mathcal{G}_k)$ is endowed with a bracket on its sheaf of sections. This bracket is known as the Spencer bracket. See [18] for a comparison of the two brackets.

[3]This is the usual definition of the Lie algebroid of the Lie groupoid $\mathrm{Aut}_k(M)$, and thus $J^k(TM/M) \to M$ can be seen as a realization of the Lie algebroid of $\mathrm{Aut}_k(M)$. However, the Lie algebroid bracket is somehow different from the Lie algebra structure that we find in the sections of $\mathrm{Lie}(\mathcal{G})$.

The equations defining $\mathrm{Lie}(\mathcal{G})$ can be found in terms of the differential invariants of $\mathcal{G}$

$$\mathrm{Lie}(\mathcal{G}) = \big\{ j_x X \in J(TM/M) \mid X^{(\infty)} \mathrm{Inv}_{\mathcal{D}}(\mathcal{G}) = 0 \big\}.$$

Note that if the differential invariants of $\mathcal{G}$ are preserved by $X^{(\infty)}$ then the flow of $\widetilde{X}^{(\infty)}$ in $\mathrm{Aut}(M)$ is tangent to $\mathcal{G}$.

1.2. **Finite-dimensional $\mathcal{D}$-groupoids.** In this paper, we are interested in $\mathcal{D}$-groupoids of finite dimension.

Definition 1.8. A $\mathcal{D}$-groupoid $\mathcal{G} = \{\mathcal{G}_m\}_{m\in\mathbb{N}} \subset \mathrm{Aut}(M)$ is said to be of finite dimension if its sequence of projections stabilizes at some integer $\ell \in \mathbb{N}$, meaning that for all $m \geq \ell$, the projections $\mathcal{G}_{m+1} \to \mathcal{G}_m$ are isomorphisms, in this case, we have

$$\mathcal{G} \simeq \cdots \simeq \mathcal{G}_{\ell+1} \simeq \mathcal{G}_\ell \to \mathcal{G}_{\ell-1} \to \cdots \to M \times M.$$

Remark 1.9. It is important to distinguish the stabilization step ℓ from the intrinsic order of the $\mathcal{D}$-groupoid (the order of the system of differential equations determining it). Throughout this paper, for a finite-dimensional $\mathcal{D}$-groupoid $\mathcal{G}$ of order q, we will always fix an integer $k \geq \max(\ell, q)$. This choice ensures that $\mathcal{G}$ is both completely determined by and faithfully represented by its k-th component, $\mathcal{G} \simeq \mathcal{G}_k$.

Definition 1.10. Let $\mathcal{G}$ be a finite-dimensional $\mathcal{D}$-groupoid. A point $x \in M$ is regular for $\mathcal{G}$ if there is a Zariski open neighborhood $M' \subset M$ of x such that $\mathcal{G}|_{M'} \subseteq \mathrm{Aut}(M')$ is a smooth Lie groupoid.

Given a finite-dimensional groupoid $\mathcal{G} \subset \mathrm{Aut}(M)$, the dimension of $\mathcal{G}$ over M is the dimension of the fibers of the source s, or the target t projection. We have $\dim_M(\mathcal{G}) = \dim(\mathcal{G}) - \dim(M)$. We will say that $\mathcal{G}$ is 0-dimensional or finite if $\dim_M(\mathcal{G}) = 0$.

1.2.1. *Adapted frames.* Let $\mathcal{G} \subset \mathrm{Aut}(M)$ be a finite-dimensional transitive[4] $\mathcal{D}$-groupoid. We also fix and order k large enough so that $\mathcal{G} \simeq \mathcal{G}_k$, and such that $\mathcal{G}$ is determined by its differential invariants of order $\leq k$. Without loss of generality, after restriction to a suitable open subset, we also assume that $\mathcal{G}_k$ is a Lie subgroupoid of $\mathrm{Aut}_k(M)$.

From now on, we identify $\mathcal{G}$ with its truncation to order k. However, it is important to remember that any element $\sigma_{xy} \in \mathcal{G}$ is an actual germ of local biholomorphism $\sigma_{xy} \colon (M, x) \to (M, y)$, completely determined by its k-jet $j^k_x \sigma_{xy} \in \mathcal{G}_k$.

Let us fix a point x, and lets recall that $\mathcal{G}_{xx}$ is the stabilizer group of x in $\mathcal{G}$.[5] In the transitive case, the moving frame method gives us an effective way of computing all differential invariants. Note that the group $\mathcal{G}_{xx}$ acts on the space of frames on x, $(R_k M)_x$ by composition. Let us fix a frame $j^k_0 r_x \in (R_k M)_x$. It induces isomorphisms

$$\begin{aligned} i_1 &\colon \Gamma_k \to (R_k M)_x, & j^k_0 \gamma &\mapsto j^k_0 (r_x \circ \gamma), \\ i_2 &\colon \Gamma_k \to \mathrm{Aut}_k(M)_{xx}, & j^k_0 \gamma &\mapsto j^k_x \big(r_x \circ \gamma \circ r_x^{-1} \big) \end{aligned}$$

that conjugate the action of $\mathcal{G}_{xx}$ in $(R_k M)_x$ with the action of the algebraic subgroup $\Gamma'_k = j^k_0 \big(r_x^{-1} \mathcal{G}_{xx} r_x \big) \subset \Gamma_k$ on Γ_k. Therefore, we have the existence of the quotient $V = (R_k M)_x / \mathcal{G}_{xx} \simeq \Gamma_k / \Gamma'_k$.

[4] In the sense that $\mathcal{G}_0 = M \times M$.

[5] Since $\mathcal{G}$ is transitive, all the points have isomorphic stabilizer groups.

Proposition 1.11. *The map $I\colon R_kM \to V$, $j_0^k r_y \mapsto [\sigma_{xy} \circ r_y]$, where σ_{xy} is any element of $\mathcal{G}_{xy}$, is a well-defined regular map, invariant by the action of $\mathcal{G}$, and it identifies $\mathrm{Inv}(\mathcal{G})$ with $\mathbb{C}(V)$. For any $v \in V$, the preimage $I^{-1}(\{v\})$ with its projection to M is a reduction of the bundle $R_kM \to M$ to a subgroup of Γ_k isomorphic to $\mathcal{G}_{xx}$. All these principal bundles are isomorphic.*

Definition 1.12. Let $\mathcal{G}$ be a finite-dimensional $\mathcal{D}$-groupoid such that $\mathcal{G} \hookrightarrow \mathrm{Aut}_k(M)$. Let us assume that $\mathcal{G}$ is also a smooth Lie groupoid. Let $I\colon R_kM \to V$ be a complete differential invariant map as in Proposition 1.11. Each bundle of the form $I^{-1}(\{v\}) \to M$ from $v \in V$ is called a bundle of adapted frames for $\mathcal{G}$.

Remark 1.13. From [9], it follows that this phenomenon actually represents the general situation for transitive Lie groupoids: every such groupoid arises as the groupoid of symmetries of a meromorphic geometric structure.

1.2.2. *$\boldsymbol{\mathcal{D}}$-Lie algebra and Lie algebroid.* Here we assume that $\mathcal{G}$ is of finite dimension $\dim_M \mathcal{G} = r$, and $\mathcal{G} \simeq \mathcal{G}_k$ is a smooth Lie subgroupoid of $\mathrm{Aut}_k(M)$. Therefore, we have $\mathrm{Lie}(\mathcal{G}) \simeq \mathrm{Lie}(\mathcal{G}_k) \subset J^k(TM/M)$, and then $\mathrm{Lie}(\mathcal{G}) \to M$ is a linear bundle of rank r of Lie algebras. The contact system on the jet bundle $T\big(J^kM/M\big)$ induces by restriction to $\mathrm{Lie}(\mathcal{G}) \to M$ a flat linear connection $\nabla^{\mathrm{Lie}(\mathcal{G})}$, and it is stable under the Lie bracket in the sense that the Lie bracket of any two horizontal sections is also a horizontal section.

Horizontal sections of $\nabla^{\mathrm{Lie}(\mathcal{G})}$ are of the form $jX\colon x \mapsto j_xX$, where X is a vector fields in M whose flow is in $\mathcal{G}$. We say that X is an infinitesimal generator of $\mathcal{G}$. The elements of the Lie algebra $\mathrm{Lie}(\mathcal{G})_x$ are the germs at x of the infinitesimal generators of $\mathcal{G}$.

Remark 1.14. A reader familiar with Lie algebroid perspective will notice that $\big(\mathrm{Lie}(\mathcal{G}), \nabla^{\mathrm{Lie}(\mathcal{G})}\big)$ is a locally symmetric Cartan algebroid in the sense of [2, Section 2.5] or a twisted Lie algebra action as in [4, Section 2.5].

1.2.3. *Example: Projective structures on the complex line.* Let $M = \mathbb{C}$ be the complex line, with coordinate λ. By a projective structure on some open subset $M' \subset M$ we mean a maximal atlas in M with values in the projective line $\overline{\mathbb{C}}$ such that the transition functions are Möbius transformations on their domain in $\overline{\mathbb{C}}$. The chart functions $\tau\colon \mathcal{U} \to \overline{\mathbb{C}}$ should satisfy the differential equation

$$S_\lambda(\tau) = R(\lambda), \qquad S_\lambda(\tau) = \frac{\tau_{\lambda\lambda\lambda}}{\tau_\lambda} - \frac{3}{2}\left(\frac{\tau_{\lambda\lambda}}{\tau_\lambda}\right)^2, \tag{1.1}$$

where $R(\lambda)$ is defined on M'. We say that the projective structure is meromorphic, or rational, if $R(\lambda) \in \mathbb{C}(\lambda)$ is a rational function. Equation (1.1) is equivalent to

$$S_\tau(\lambda) + R(\lambda)\lambda_\tau^2 = 0. \tag{1.2}$$

Now, let us consider $\mathrm{Aut}_3(\mathbb{C})$, with coordinates λ, φ, φ_λ, $\varphi_{\lambda\lambda}$, $\varphi_{\lambda\lambda\lambda}$. Let us derive the equations for the symmetries φ of (1.2), in the sense that $\lambda(\tau)$ is a solution of (1.2), if and only if $\varphi(\lambda(\tau))$ is also a solution. By the chain rule for the Schwarzian derivative, we obtain

$$S_\lambda(\varphi) + \varphi_\lambda^2 R(\varphi) = R(\lambda). \tag{1.3}$$

This is the equation of a groupoid $\mathcal{G}_3 \subset \mathrm{Aut}_3(\mathbb{C})$. Let us compute the third-order differential invariant giving rise to the adapted frame bundles.

Let us consider a frame $(\lambda, \lambda_\varepsilon, \lambda_{\varepsilon\varepsilon}, \lambda_{\varepsilon\varepsilon\varepsilon})$. If we consider a 3-jet of biholomorphism $(\lambda, \varphi, \varphi_\lambda, \varphi_{\lambda\lambda}, \varphi_{\lambda\lambda\lambda})$, the composition gives us a new frame $(\varphi, \varphi_\varepsilon, \varphi_{\varepsilon\varepsilon}, \varphi_{\varepsilon\varepsilon\varepsilon})$ where, as before for the chain rule $S_\varepsilon(\varphi) = S_\lambda(\varphi)\lambda_\varepsilon^2 + S_\varepsilon(\lambda)$, and replacing $S_\lambda(\varphi)$ from the groupoid equation (1.3), we obtain

$$S_\varepsilon(\varphi) + R(\varphi)\varphi_\varepsilon^2 = S_\varepsilon(\lambda) + R(\lambda)\lambda_\varepsilon^2,$$

and therefore

$$I(\lambda, \lambda_\varepsilon, \lambda_{\varepsilon\varepsilon}, \lambda_{\varepsilon\varepsilon\varepsilon}) = R(\lambda)\lambda_\varepsilon^2 + \frac{\lambda_{\varepsilon\varepsilon\varepsilon}}{\lambda_\varepsilon} - \frac{3}{2}\left(\frac{\lambda_{\varepsilon\varepsilon}}{\lambda_\varepsilon}\right)^2 \tag{1.4}$$

is a differential invariant for the groupoid $\mathcal{G}_3$. Let us compute the equation for the Lie algebra of $\mathcal{G}_3$. Let $X = a(\lambda)\frac{\partial}{\partial\lambda}$ be a vector field in $\mathbb{C}$. Its third-order prolongation $X^{(3)}$ is then

$$X^{(3)} = a\frac{\partial}{\partial\lambda} + a_\lambda\lambda_\varepsilon\frac{\partial}{\partial\lambda_\varepsilon} + \big(a_{\lambda\lambda}\lambda_\varepsilon^2 + a_\lambda\lambda_{\varepsilon\varepsilon}\big)\frac{\partial}{\partial\lambda_{\varepsilon\varepsilon}} + \big(a_{\lambda\lambda\lambda}\lambda_\varepsilon^3 + 3a_{\lambda\lambda}\lambda_\varepsilon\lambda_{\varepsilon\varepsilon} + a_\lambda\lambda_{\varepsilon\varepsilon\varepsilon}\big)\frac{\partial}{\partial\lambda_{\varepsilon\varepsilon\varepsilon}},$$

and a direct computation gives us

$$X^{(3)}I = (a_{\lambda\lambda\lambda} + 2R(\lambda)a_\lambda + R'(\lambda)a)\lambda_\varepsilon^2.$$

So that, we find the linear differential equation defining the Lie algebra of $\mathcal{G}_3$,

$$a_{\lambda\lambda\lambda} + 2R(\lambda)a_\lambda + R'(\lambda)a = 0. \tag{1.5}$$

1.3. **Picard–Vessiot theory.** There are several presentations of Picard–Vessiot theory, also known as linear differential Galois theory. In [5, Appendix A], we provided a geometrical presentation that is equivalent to Kolchin's theory of strongly normal extensions of finite transcendence degree over $\mathbb{C}$. We recall here the main points of the theory in a simplified way, that applies to the examples and applications developed in this paper. We direct the reader to the aforementioned reference for a full exposition.

1.3.1. *Definition of the Galois group.* Let us consider G an algebraic group, $\pi\colon P \to M$ a connected principal bundle modeled over G, and a flat rational connection with connection form $\theta\colon P \dashrightarrow \mathrm{Lie}(G)$. Let us call M' a Zariski open subset of M such that θ is regular on $P|_{M'} = P'$. Then $\ker(\theta) = \mathcal{F} \subset TP'$ is a foliation on P'. For a given point $p \in P'$ let $Z_p = \overline{\mathcal{L}_p}$ be the Zariski closure of the leave of $\mathcal{F}$ passing through p. Then, we can define a subgroup of G,

$$\mathrm{Gal}(\theta, p) = \{g \in G \mid Z_p g = Z_p\} \subseteq G,$$

which is called the Galois group of the connection given by θ at the point p. It has the following properties:

(1) $\mathrm{Gal}(\theta, p)$ is an algebraic subgroup of G.
(2) $Z_p \to M'$ is an irreducible principal bundle modeled over $\mathrm{Gal}(\theta, p)$.
(3) All Galois groups are isomorphic and belong to the same conjugation class as subgroups of G.

Thus, we may write $\mathrm{Gal}(\theta)$ to refer to the Galois group, as an abstract algebraic group independent of the point p in P'. We also write $\mathfrak{gal}(\theta)$ to refer to the Lie algebra of the Galois group. Note that $\mathfrak{gal}(\theta) = 0$ if and only if the Galois group is finite.

1.3.2. *Galois group for linear connections.* The above construction applies to linear connections in the following way. Let us consider a linear bundle $E \to M$ of rank k, and ∇ a flat linear rational connection. As previously, we consider M' the domain of the connection and E' the restriction of E to M', so that, ∇ is a regular connection over M'. Let E_x be the fiber at $x \in M'$. The bundle $P \subset E_x^* \otimes E$ whose fiber at $x' \in M'$ is the set of linear isomorphisms from E_x to $E_{x'}$ is a $\mathrm{GL}(E_x)$-principal bundle over M'. The connection ∇ induces a principal connection with connection form

$$\theta_\nabla(X_p) = p^{-1} \circ \nabla_{\pi_*(X_p)}p \in \mathfrak{gl}(V),$$

where $p\colon E_x \to E_{x'}$ is a linear isomorphism and $\pi_*(X_p)$ is the projection of the vector $X_p \in T_p(P)$ on M'. Now P has a special point, $\mathrm{id}\colon E_x \to E_x$ and $\mathrm{Gal}(\theta_\nabla, \mathrm{id})$,

denoted by $\mathrm{Gal}(\nabla, x)$, is the Galois group of ∇ at x. Note that $\mathrm{Gal}(\nabla, x) \subset \mathrm{GL}(E_x)$, so that the space E_x is naturally endowed with a representation of the Galois group $\mathrm{Gal}(\nabla, x)$.

Remark 1.15. An important property of this construction is that it is compatible with direct product. If $E_i \to M_i$ with $i = 1, \ldots, k$ is a finite family of vector bundles endowed with flat rational linear connections ∇_i, then we may consider the direct product connection $\nabla = \prod \nabla_i$ in the direct product bundle $E = \prod E_i \to M = \prod M_i$, and in such case $\mathrm{Gal}(\nabla, (x_1, \ldots, x_k)) = \prod \mathrm{Gal}(\nabla_i, x_i)$, acting on $E_{(x_1,\ldots,x_k)} = \prod (E_i)_{x_i}$

1.4. **Parallelisms.** This section is a recap of the main concepts and results presented in [5]. Let $\mathfrak{g}$ be a complex Lie algebra, and M a smooth complex irreducible algebraic variety over $\mathbb{C}$ with $\dim M = n$.

Definition 1.16. A $\mathfrak{g}$-*parallelism* in M is a Lie algebra morphism $\varphi\colon \mathfrak{g} \to \mathfrak{X}(M)$ such that for all x in M the linear map $\mathrm{ev}_x \circ \varphi\colon \mathfrak{g} \to T_xM$, $A \mapsto \varphi(A)_x$ is an isomorphism. The image of $\varphi(\mathfrak{g}) \subset \mathfrak{X}(M)$ is the Lie algebra of parallel vector fields of φ. A parallelism in M is a $\mathfrak{g}$-parallelism for some Lie algebra $\mathfrak{g}$.

By duality, any $\mathfrak{g}$-parallelism can be codified into a $\mathfrak{g}$-*parallelism form* $\omega\colon TM \to \mathfrak{g}$ by defining $\omega(X_x) = (\mathrm{ev}_x \circ \varphi)^{-1}(X_x)$. It is possible to give the following definition. It follows that any $\mathfrak{g}$-valued 1-form ω in M is a parallelism form if and only if it satisfies

(1) For all $x \in M$, $\omega_x\colon T_xM \to \mathfrak{g}$ is a linear isomorphism.
(2) If A, B are elements of $\mathfrak{g}$ and X, Y are the vector fields in M such that $\omega(X) = A$ and $\omega(Y) = B$, then $\omega([X, Y]) = [A, B]$, or equivalently:
(2') It satisfies the Maurer–Cartan equation $\mathrm{d}\omega + \frac{1}{2}[\omega, \omega] = 0$.

With this dual definition, a parallel vector field with respect the parallelism form ω is a vector field X with constant $\omega(X) \in \mathfrak{g}$. A $\mathfrak{g}$-parallelism in M is considered to be regular, rational, analytic, formal, etc., depending on the degree of regularity of its corresponding parallelism form. A $\mathfrak{g}$-*parallelized variety* is then a pair (M, ω), where ω is a $\mathfrak{g}$-parallelism form, and a conjugation of parallelized varieties $f\colon (M, \omega) \to (N, \varpi)$ is a map $f\colon N \to M$ such that $f^*(\varpi) = \omega$. As before we may consider regular, rational, analytic or formal morphisms.

The main example of parallelized variety is (G, ω_G), where G is an algebraic group and $\omega_G\colon TG \to \mathrm{Lie}(G) = T_eG$ is its structure for $\omega_G(X_g) = \mathrm{d}L_g^{-1}(X_g)$. Parallel vector fields of ω_G are left invariant vector fields. Note that there is a reciprocal parallelism defined by $\omega_G^{\mathrm{rec}}(X_g) = -\mathrm{d}R_g^{-1}(X_g)$, whose parallel vector fields are the right invariant vector fields in G. Note that the inversion $(G, \omega_G) \to \left(G, \omega_G^{\mathrm{rec}}\right)$, $g \mapsto g^{-1}$ is a regular conjugation between the parallelisms ω_G and ω_G^{rec}.

Given varieties (M, ω) and (N, ϖ) with rational parallelisms, we are interested in studying birational conjugations, but also multivalued algebraic conjugations $f\colon M \dashrightarrow N$. A way of understanding these multivalued algebraic maps is through the notion of *isogeny*. We say that (M, ω) and (N, ϖ) are isogenous if there is an irreducible parallelized variety (U, θ) and dominant maps $\pi_M\colon U \to M$, $\pi_N\colon U \to N$ such that $\pi_M^*(\omega) = \pi_N^*(\varpi) = \theta$. We may assume that U is a covering of some open subvarieties of M and N. This notion extends the classical notion of isogeny from algebraic groups.

Example 1.17. Let (G, ω_G) be a connected algebraic group with its structure form. Let $H \subset G$ be a finite group, so that the form ω_G is projectable to the quotient

$$\pi\colon M = G \to H\backslash G = \{Hg \mid g \in G\}$$

then $(M, \pi_*(\omega_G))$ is isogenous to (G, ω_G). Moreover, let $f\colon N \dashrightarrow M$ be a dominant map, with N irreducible variety of the same dimension as M. Then $\big(N, f^*(\pi_*(\omega_G))\big)$ is isogenous to (G, ω_G).

1.5. **Lie connections.** In this section, we deal with affine connections in M, that is, linear connections on $TM \to M$. First, let us note that given an affine connection ∇ we can always define a reciprocal connection by the formula $\nabla^{\mathrm{rec}}_X Y = \nabla_Y X + [X, Y]$. From this definition, the difference $\nabla - \nabla^{\mathrm{rec}} = \mathrm{Tor}_\nabla$ is the torsion tensor. Let us now consider the following definition, taken from [5].

Definition 1.18. A Lie connection in M is a flat connection ∇ in TM such that the Lie bracket of two horizontal vector fields is also horizontal.

In other words, a Lie connection is a linear differential equation that locally defines a parallelism. We may say that a Lie connection is modeled over the Lie algebra $\mathfrak{g}$ if the space of horizontal vector fields defined around any regular point is a Lie algebra isomorphic to $\mathfrak{g}$. Lie connections can be characterized in the following way, as is done in [5, Proposition 3.10].

Proposition 1.19. *Let ∇ be a Lie connection in TM, the following statements are equivalent:*

1. *∇ is a Lie connection.*
2. *∇^{rec} is a Lie connection.*
3. *∇ is flat and has constant torsion in the sense $\nabla \mathrm{Tor}_\nabla = 0$.*
4. *∇ and ∇^{rec} are flat.*

There is an interesting relationship between a pair of reciprocal Lie connections ∇ and ∇^{rec}. They are modeled over the same Lie algebra. Horizontal vector fields for ∇ commute with the horizontal vector fields of ∇^{rec} and vice-versa. We are in an infinitesimal situation that is analogous to that of the Lie algebras of left and right invariant vector fields in an algebraic group.

Given a $\mathfrak{g}$-parallelism form ω, we can always find the unique Lie connection ∇ whose horizontal vector fields are those that are parallel with respect to ω. Given a basis $\{A_1, \ldots, A_n\}$, we can fix a frame $(X_1, \ldots, X_r)$ in M defined by $\omega(X_i) = A_i$, and then we define ∇ such that $\nabla_{X_i} X_j = 0$ for all $1 \leq i,\, j \leq n$. Then ∇^{rec} is also a Lie connection modeled over the same Lie algebra $\mathfrak{g}$ and its horizontal vector fields are the infinitesimal symmetries of ω, in the sense that $\nabla^{\mathrm{rec}} Y = 0$ if and only if $\mathcal{L}_Y \omega = 0$.

Theorem 1.20 ([5, Corollary 4.3 (a) and (d)])**.** *Let $\mathfrak{g}$ be a centerless Lie algebra. An algebraic variety (M, ω) with a rational parallelism of type $\mathfrak{g}$ is isogenous to an algebraic group if and only if $\mathfrak{gal}\big(\nabla^{\mathrm{rec}}\big) = \{0\}$. Moreover, if $\mathrm{Gal}\big(\nabla^{\mathrm{rec}}\big) = \{\mathrm{id}\}$ then the isogeny is given by a dominant rational map $\pi\colon M \dashrightarrow G$ such that $\pi^*(\omega_G) = \omega$.*

1.6. **Minimality and n-minimality.** There is a natural embedding of $\mathrm{Aut}(M) \times \mathrm{Aut}(M) \subset \mathrm{Aut}(M \times M)$, $\mathrm{Aut}(M) \times \mathrm{Aut}(M)$ is the first-order $\mathcal{D}$-groupoid on $M \times M$ defined by equations, $\frac{\partial \sigma_1}{\partial x_2} = \frac{\partial \sigma_2}{\partial x_1} = 0$ for $\sigma = (\sigma_1, \sigma_2)\colon M \times M \to M \times M$, $(x_1, x_2) \mapsto (\sigma_1(x_1, x_2), \sigma_2(x_1, x_2))$. However, it was first noted by Sophus Lie that the diagonal action of $\mathrm{Aut}(M)$ on M, that is, the equation $\sigma_1 = \sigma_2$ can not be defined by using differential equations. There is a diagonal pseudogroup of $\mathrm{Aut}(M) \times \mathrm{Aut}(M)$ acting on $M \times M$, but it is not a $\mathcal{D}$-groupoid (or Lie pseudogroup). In broad terms, $\mathcal{D}$-groupoids may admit fewer sub-$\mathcal{D}$-groupoids than algebraic groups admit algebraic subgroups.

If $\mathcal{G}$ is isogenous to an action groupoid $G \ltimes M$, then clearly the diagonal of $\mathcal{G} \times \mathcal{G}$ is a sub-$\mathcal{D}$-groupoid isogenous to the diagonal of G acting on $M \times M$. This motivates the following definition.

Definition 1.21. A $\mathcal{D}$-groupoid $\mathcal{G}$ on M is n-minimal if for any sub-$\mathcal{D}$-groupoid $\mathcal{H} \subset \mathcal{G}^n$, if $\dim_{M^n}(\mathcal{H}) < \dim_{M^n}\big(\mathcal{G}^n\big)$, then there exists i such that $\dim_M(\mathrm{pr}_i(\mathcal{H})) = 0$, where $\mathrm{pr}_i\colon \mathcal{G}^n \to \mathcal{G}$ is the projection on the i-th factor.

Remark 1.22. We will say minimal for 1-minimal. That is, $\mathcal{G}$ is minimal if it does not admit nontrivial sub-$\mathcal{D}$-groupoids of positive dimension.

Example 1.23. First examples of minimal $\mathcal{D}$-groupoid were given in [10, Proposition 3.6], where if is shown that the Kummer groupoid of symmetries of a projective structure (1.3) is minimal if and only if the Ricatti equation

$$u_\lambda + u^2 + \frac{1}{2}R(\lambda) = 0 \tag{1.6}$$

has no solutions in $\mathbb{C}(\lambda)^{\mathrm{alg}}$, the algebraic closure of $\mathbb{C}(\lambda)$. This relation between (1.6) and the third-order linear equation (1.5) comes through the second-order differential equation

$$\psi_{\lambda\lambda} + \frac{1}{2}R(\lambda)\psi = 0. \tag{1.7}$$

On one hand, logarithmic derivatives $u = \psi_\lambda/\psi$ of solutions of (1.7) satisfy the differential equation (1.6). On the other hand, (1.5) is the second symmetric power of (1.7) in the sense that for any basis of solutions ψ_1, ψ_2 of (1.7) the quadratic products $a_1 = \psi_1^2$, $a_2 = \psi_1\psi_2$, $a_3 = \psi_2^2$ form a basis of solutions of (1.5).

Let us consider $x \in \mathbb{C}$ a regular point for R, the Galois group $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big)$ acts on the space $\mathrm{Lie}(\mathcal{G})_x \simeq \mathfrak{sl}_2$ by automorphisms of Lie algebras. Then we have

$$\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big) \subset \mathrm{Aut}(\mathrm{Lie}(\mathcal{G})_x) \simeq \mathrm{PSL}_2.$$

On one hand, any nontrivial positive-dimensional sub-$\mathcal{D}$-groupoid of $\mathcal{G}$ has a nontrivial sub-$\mathcal{D}$-algebra of $\mathrm{Lie}(\mathcal{G})$. Such a nontrivial subalgebra of $\mathrm{Lie}(\mathcal{G})$ will give us a nontrivial $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big)$-invariant Lie subalgebra of $\mathrm{Lie}(\mathcal{G})_x$, and then the reduction of the action of the Galois group on the space of initial conditions.

Reciprocally, the realization of (1.5) as the symmetric square of (1.7) informs us that any $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big)$-invariant subspace of $\mathrm{Lie}(\mathcal{G})_x$ also implies that the Galois group of (1.7) is inside a Borel subgroup of SL_2. In such case, (1.6) has a rational solution.

This rational solution u can be used to find a proper sub-$\mathcal{D}$-groupoid in the following way. Take $r = -2u \to r' = \frac{1}{2}r^2 + R$, and define

$$\mathcal{B} = \{j_x(a\partial_\lambda) \in J(T\mathbb{C}/\mathbb{C}) \mid a_{\lambda\lambda} + ra_\lambda + r_\lambda a = 0\}.$$

Then $\mathcal{B}$ is a 2-dimensional subalgebra of $\mathrm{Lie}(\mathcal{G})$ as it follows from $R = r' - \frac{1}{2}r^2$. It is a Lie $\mathcal{D}$-algebra of affine symmetries of dimension 2, sitting inside the 3-dimensional projective symmetry algebra $\mathrm{Lie}(\mathcal{G})$.

Consider the transformation rule $r^* = \frac{\varphi_{\lambda\lambda}}{\varphi_\lambda} + \varphi_\lambda r(\varphi)$. Differentiate at $t = 0$ along a flow φ_t with generator $a(\lambda)\partial_\lambda$

$$\delta r := \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} r^* = a_{\lambda\lambda} + ra_\lambda + r_\lambda a,$$

so $a \in \mathcal{B} \iff \delta r = 0 \iff$ the flow preserves the affine connection r and the set of diffeomorphisms preserving r is automatically a nontrivial 2-dimensional subpseudogroup of $\mathcal{G}$. Then we have the equivalence between the following conditions:

(i) The associated Riccati equation (1.6) has no rational solution.
(ii) There is no nontrivial $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big)$-invariant subspace in $\mathrm{Lie}(\mathcal{G})_x$.
(iii) There is no nontrivial $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big)$-invariant Lie subalgebra in $\mathrm{Lie}(\mathcal{G})_x$.
(iv) $\mathcal{G}$ is minimal.

Remark 1.24. The reader may note that in [10, Proposition 3.6] it is stated that the existence of an algebraic but not rational solution of the Ricatti equation (1.6) also implies the existence of a proper sub-$\mathcal{D}$-groupoid of $\mathcal{G}$. However, such a $\mathcal{D}$-groupoid is not a $\mathcal{D}$-groupoid $\mathcal{H} \subset \mathcal{G} \subset \mathrm{Aut}(\mathbb{C})$. Such a groupoid is of the form $\mathcal{H} \subset \pi^*(\mathcal{G}) \subset \mathrm{Aut}(M)$ where $\pi\colon M \to \mathbb{C}$ is the algebraic curve associated to the algebraic solution of (1.6). Therefore, it is not considered here as a proper sub-$\mathcal{D}$-groupoid of $\mathcal{G}$.

Definition 1.25. For a Lie algebra $\mathfrak{g}$, a subgroup of $\mathrm{Aut}(\mathfrak{g})$ is Lie-irreducible if there is no invariant proper Lie subalgebra.

Theorem 1.26. *Let $\mathcal{G}$ be a finite-dimensional transitive $\mathcal{D}$-groupoid on M, let $x \in M$ be a regular point for $\mathcal{G}$, and let $\mathrm{Lie}(\mathcal{G})_x$ be the Lie algebra at x. If $\mathrm{Gal}(\mathrm{Lie}(\mathcal{G}), x) \subset \mathrm{Aut}(\mathrm{Lie}(\mathcal{G})_x)$ is Lie-irreducible, then for any $n \in \mathbb{N}$, $\mathcal{G}$ is n-minimal.*

Proof. Let us first prove the minimality (= 1-minimality). If $\mathcal{H} \subsetneq \mathcal{G}$ is such that $\dim_M \mathcal{H} < \dim_M \mathcal{G}$, then $\mathrm{Lie}(\mathcal{H}) \subsetneq \mathrm{Lie}(\mathcal{G})$ is a vector sub-bundle invariant under the connection. Then the fiber at x: $\mathrm{Lie}(\mathcal{H})_x \subsetneq \mathrm{Lie}(\mathcal{G})_x$ is a Lie subalgebra invariant under the Galois group $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big)$. Under our hypothesis, it is $\{0\}$ meaning that $\dim_M \mathcal{H} = 0$.

Now assume $n > 0$. Let $\mathcal{H} \subsetneq \mathcal{G}^n$ such that $\dim_{M^n} \mathcal{H} < \dim_{M^n} \mathcal{G}^n$. Since

$$\mathrm{Gal}\big(\mathrm{Lie}(\mathcal{G})^n, (x_1, \dots, x_n)\big) = \prod \mathrm{Gal}(\mathrm{Lie}(\mathcal{G}), x_i),$$

the possible proper Lie algebras

$$\mathrm{Lie}(\mathcal{H})_{(x_1,\dots,x_n)} \subset \prod \mathrm{Lie}(\mathcal{G})_{x_i}$$

invariant under $\mathrm{Gal}\big(\mathrm{Lie}(\mathcal{G})^n, (x_1, \dots, x_n)\big)$ are the products of factors: $\mathrm{Lie}(\mathcal{H})_{(x_1,\dots,x_n)} = \prod \mathfrak{h}_i$. Then there exists i such that $\mathrm{pr}_i(\mathrm{Lie}(\mathcal{H})_{(x_1,\dots,x_n)})$ is a proper Lie subalgebra of $\mathrm{Lie}(\mathcal{G})_{x_i}$ invariant under $\mathrm{Gal}(\mathrm{Lie}(\mathcal{G}), x_i)$. By Lie-irreducibility it is $\{0\}$.

This implies that if $\mathcal{H} \subset \mathcal{G}^n$ is such that $\dim_{M^n}(\mathcal{H}) < \dim_{M^n}\big(\mathcal{G}^n\big)$ then its projection on the i-th factor is 0-dimensional. □

By simple Lie algebra, we mean a finite-dimensional noncommutative Lie algebra without proper ideals.

Theorem 1.27. *Let $\mathcal{G}$ be a finite-dimensional transitive $\mathcal{D}$-groupoid on M, let $x \in M$ be a regular point of $\mathcal{G}$, and let $\mathrm{Lie}(\mathcal{G})_x$ be the Lie algebra at x. Assume $\mathrm{Lie}(\mathcal{G})_x$ is simple and $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x\big) \neq \{\mathrm{id}\}$. For any $n > 1$ there exists no proper sub-$\mathcal{D}$-groupoid $\mathcal{H} \subset \mathcal{G}^n$ whose projections on each factor are onto.*

Proof. Let $\mathcal{H} \subset \mathcal{G}^n$ be a proper subgroupoid with projections on each factor onto. Let $\mathrm{Lie}(\mathcal{H})_{(x_1,\dots,x_n)} \subset \prod \mathrm{Lie}(\mathcal{G})_{x_i}$ be its Lie algebra at a generic regular point $(x_1, \dots, x_n) \in M^n$. As the projections on each factor are onto, by Goursat's lemma, there exists $\ell < k$ such that the projection onto the (ℓ, k)-th factors of $\mathrm{Lie}(\mathcal{H})_{(x_1,\dots,x_n)}$ is the graph of an isomorphism $\mathrm{Lie}(\mathcal{G})_{x_\ell} \to \mathrm{Lie}(\mathcal{G})_{x_k}$.

As $\mathrm{Lie}(\mathcal{H}) \subset \mathrm{Lie}(\mathcal{G})^n$ is a $\nabla^{\mathrm{Lie}(\mathcal{G})^n}$-invariant sub-bundle, then its fiber $\mathrm{Lie}(\mathcal{H})_{(x_1,\dots,x_n)}$ must be invariant under the action of $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})^n}, (x_1, \dots, x_n)\big)$. This implies that the projection must be $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x_\ell\big) \times \mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x_k\big)$ invariant. This implies $\mathrm{Gal}\big(\nabla^{\mathrm{Lie}(\mathcal{G})}, x_\ell\big) = \{\mathrm{id}\}$. □

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